

Aayush Gnawali

AI • Backend • DevOps, Kathmandu, Nepal

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Summary

Software engineer with hands-on experience in backend development, AI/ML systems, and DevOps practices. Skilled in building reliable APIs and data-driven applications, developing practical AI and RAG workflows, containerizing services, and supporting automated deployment pipelines. Brings a strong focus on clean architecture, secure data handling, maintainable code, and dependable production operations. Seeking opportunities in backend engineering, AI engineering, DevOps, or related software roles, and open to on-site, hybrid, and remote work.

Experience

DevOps Intern

2026

Lego Tech Pvt. Ltd.

Remote

- Applied Linux fundamentals and Bash scripting to support day-to-day system and development workflows.
- Built and managed containerized services using Docker and Docker Compose with YAML-driven configuration and deployment tasks.
- Gained hands-on experience with AWS core services, including EC2 for infrastructure support and deployment basics.
- Learned CI/CD practices, deployment fundamentals, and DevOps workflows in a professional distributed engineering environment.
- Worked with Git for version control and collaborated across infrastructure and application-related tasks.

AI & Backend Intern

May 2026 – August 2026

E-Digital Nepal Pvt. Ltd.

On-site

- Developed AI assistant features, document ingestion pipelines, semantic search, knowledge base integrations, and Retrieval-Augmented Generation (RAG) workflows for education and school-management platforms.
- Built backend services using Python and PostgreSQL for APIs, authentication, role-based access control, and secure data management.
- Contributed to admin tools, dynamic form management, user tracking, monitoring, and platform reliability for school management systems.
- Maintained secure data separation and production-oriented system practices across organizational data environments.
- Collaborated with the engineering team to improve conversational workflows and backend features for real-world platform usage.

Projects

Portfolio RAG Assistant

Live Demo

- Built a portfolio chatbot that retrieves evidence from structured portfolio content, chunks and embeds it with a vector database, and answers questions with grounded, citation-aware responses.
- Implemented a server-side retrieval flow with public-only document filtering, prompt isolation, and validation of LLM output to reduce hallucinations and keep answers grounded in verified

portfolio facts.

- Designed the assistant to abstain when evidence is insufficient, demonstrating practical AI safety and real-world production thinking.

FlipClock

Live Demo

- Built a customizable Windows desktop clock overlay with animated split-flap digits, four display themes, system-tray controls, persistent user settings, and multi-monitor positioning.
- Implemented Windows startup support, single-instance protection, opacity controls, wallpaper mode, and multi-monitor position recovery.
- Automated executable packaging and installer release workflows using PyInstaller, Inno Setup, and GitHub Actions.

Rule-Based Self-Healing Web Automation System

- Developed an automation system that detects UI drift and recovers from failed browser workflows using rule-based healing logic instead of manual intervention.
- Designed resilient browser-state tracking and recovery steps to handle selector changes and dynamic app updates.

House Price Prediction Deployment

Live Demo

- Built and deployed a machine learning regression app that turned a notebook workflow into a live, interactive prediction interface.
- Covered data preprocessing, feature engineering, and real-time inference with model monitoring and user-facing visualizations.

HippyHub E-Commerce Website

GitHub Repository

- Built a full-stack e-commerce platform for hemp products and Dhaka-pattern clothing using React on the frontend and FastAPI with PostgreSQL on the backend.
- Containerized the stack with Docker and Docker Compose and learned production-oriented deployment and service orchestration.

Skills

Backend: FastAPI, REST APIs, Async SQLAlchemy, Pydantic, JWT Authentication, OOP

DevOps: Docker, Docker Compose, Linux, Bash, YAML, CI/CD Concepts, Git, GitHub

AI / ML: PyTorch, Scikit-learn, Pandas, NumPy, Model Deployment

Databases: PostgreSQL, MySQL

Cloud / Deployment: Render, Streamlit Cloud, GitHub Pages, Deployment Fundamentals, AWS

Tools: Git, GitHub, Selenium, Streamlit, Virtual Environments, Figma

Certifications

- **AI/ML Workshop** – ITSNP, 2025
Intensive training on neural network architectures and implementation, including CNNs.
- **Robotics Training** – Everest Engineering College, 2025
Embedded systems and automation fundamentals.
- **DevOps and Cloud Computing Training** – Upskill Nepal, 2026
Hands-on cloud and DevOps training covering deployment, automation, and infrastructure best practices.

Education

B.E. in Information Technology

2022 – Present

Everest Engineering College (Pokhara University)

8th Semester

Achievements

- **Chief Content Writer** – Hult Prize Organizing Committee, Everest Engineering College, 2025
Contributed to strategic communication and event content planning; the team ranked in the Top 25% in South Asia among roughly 300 organizing committees.
- **Best Minor Project** – Everest Engineering College, 2025
Won best minor project among all 6th semester submissions for technical execution, problem solving, and presentation.